DEGRACE companions in MG-GLAM: status

What landed, how it was verified, what the first pilot says, what is left

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17 October 2026 merged into `cz`; plan in `docs/`

One week from plan to merged code

- **Scope** (plan of 16 Oct): 42 exact outputs per box (17 primary redshifts + 25 stations at $\Delta\chi = 256\,h^{-1}$Mpc), lens planes instead of snapshots, and the extra products of the brief: δ_L, low-k density and momentum modes, a hashed subsample, f_R at halo centres.
- **Implemented** on `degrace-track-a` (17 commits, 53 files, +10.8k lines): items I1–I10; I11 (optional) not done.
- **Verified**: 180-check regression suite + four unit/integration scripts, six independent code reviews (0 blockers, 8 majors, all fixed), two pilots on ECOSMOG node 2.
- **Merged** into `cz` today (fast-forward, suite green on `cz`).

Take-away

Everything a companion run needs exists, is tested and is on the mainline. Three design decisions and the f(R) pilot verification remain.

Status of the plan items

item	what	status	evidence
I1	exact-z scheduler accepts arbitrarily close output pairs	**done** (`cz` ae4fa9d)	suite case 7
I2	lens planes: 12 per output, $\Sigma/\bar{\Sigma}$ on 2048^2	**done**	L1–L3
I3–I5	input generator, `OutputSchedule.dat`, per-output classes	**done**	cases 9, 4e; S2 (42 epochs = ECOSMOG to 10^{-7})
I6	provenance per box	partial	JSON lacks generator/LPT/seed (minor)
I7	$\delta_L(k)$ dump, route 1: in our FML IC generator	**done**	V3, V7, V8; `run_deltaL_test.sh`
I8	low-k δ_m and $(1+\delta)$v modes	**done**	V1, V2, mode-by-mode
I9	ID-hashed quantised subsample, 1 in M	**done**	V5, V6; `run_ic2pm_test.sh`
I10	f_R at BDM halo centres	**done**	unit test + f(R) smoke case
I11	optional extras	**open**	not started
pilot	node 2 ΛCDM, $N_g = 2048$	**done**	verified, 0 FAIL
pilot	node 2 f(R), extras + subsample + f_R	**running**	job 12180110, $\sim$7 h

All flags default off: without the new `Init.dat` lines the code follows the `cz` path (review 2), apart from `OutputSchedule.dat` and an appended `Setup.dat` block.

Schedule and output classes (l1, l4, l5)

- `#outputs = -42`: every requested epoch is hit exactly (a step moved onto it or inserted); pairs closer than 0.5 % allowed (four in the DEGRACE list).
- Pilot: 181 steps = 163 ladder + 18 inserted, $\mathrm{d}a_0 = 6 \times 10^{-4}$ from $z = 49$.
- `OutputSchedule.dat`: step, a, z, χ, exact flag, class.
- Class 1 = full (halo finder + extras), 2 = station (planes + $P(k)$). Production decision: extras at all 42 outputs.

step	a	z	χ	exact	class
57	0.10275	8.732	8344.5	1	1
63	0.11832	7.452	8104.7	1	2
105	0.33333	2.000	5190.2	1	1
127	0.50000	1.000	3306.9	1	1
181	1.00000	0.000	0.0	1	1

χ in h^{-1}Mpc. Epochs equal the pinned ECOSMOG list of node 2 to 10^{-7}.

Lens planes (l2): cheap, verified, 1 % of the run

- Per output: 3 axes $\times$ 4 slabs of $256\,h^{-1}$Mpc, mean of $1+\delta$ over the slab column, block-averaged to 2048^2; float32, 201 MB per output, 8.5 GB per box.
- `.meta` sidecar: geometry, epoch, pixel convention, cosmology, header.
- Cost on the pilot: 1.8 s per output; the 201 MB write 12 s.

check	result
L1 mass conservation per axis	10^{-7}
L2 vs `ProjectDENSIT`	10^{-5}
L2 36^2 = block mean of 72^2	10^{-5}
L3 P_{2D} vs projected P_{3D}	3σ per ring; 3–7 %

Open question for Baojiu

Plane normalisation ($\Sigma/\bar{\Sigma}$ vs Σ); do the stations also need halo catalogues?

Fourier dumps, subsample, f_R at haloes (l8–l10)

product	what	verified
`deltam`	$\delta_k = N^{-3}\sum_x \delta(x)e^{-ik\cdot x}$, cube $\lvert k_i\rvert \le k_c = 1.5$, CIC on N_g, complex64 $C(0{:}n, -n{:}n, -n{:}n)$	V1 layout; V2 = `PowerDM.R` to 10^{-4}
`mom_x,y,z`	$(1+\delta)\,v$ on $N_{\rm dep} = N_g/2$, v at $a_v = a - \mathrm{d}a/2$, $k_c = 0.75$ (below $k_{N,\rm dep}/4$: no interlacing)	V6; mode by mode vs decoded particles 7×10^{-5}
`Subsample`	keep 1 in M by splitmix64(ID $\oplus$ salt), int16 positions and velocities, deterministic across twins, threads, restarts	V5 (10^7 IDs), V6
`fR_halo`	$f_R = f_{R,\rm bg}(a_{\rm old})\,u^{n+1}$ at every BDM centre	unit test; f(R) smoke case

- Storage at 1024^3 per output: δ_m 0.47 GB, momentum 0.41 GB, subsample ($M = 100$) 0.13 GB.
- From the review: `ic2pm` places every particle at array index = Gadget ID and tags the lattice order (`IDz` 2LPTic, `IDx` FML) in the PM header, so the subsample index gives the Lagrangian position.

δ_L from the IC generator (I7, route 1)

- The FML generator writes `<prefix>_deltaL.bin/.meta`: its Gaussian field at $z_{\rm ini}$ divided by $D(a_{\rm ini})$, same layout and convention as `deltam`; $k_c = 1.5$, 0.47 GB.
- Lua: `write_deltaL`, `deltaL_kc`; on in the production input. Costs 1 s.
- Baojiu's ICs and the legacy pilots are out of scope (decision of 17 Oct).

check	result
Zel'dovich: $\delta_L = -ik\cdot\Psi(k)/D$	every mode to 10^{-3}
2LPT ICs	residual 8×10^{-3} (2nd order)
wrong axis / sign / half cell	off by $10^4\times$
V3: ratio to $D(a)$, $k \le 0.15$	0.9997–1.0012
V3: cross-correlation, $k \le 0.05$	0.99998
V3: coordinate offset per axis	$\le 0.03\,h^{-1}$Mpc
V7: 1/2/4 MPI ranks	identical md5

Lesson

CIC density of particles sitting on grid nodes is not linear in the displacement (phase scatter $\sim k\Delta/2$ per mode at early times): compare dumps with binned means and cross-correlations.

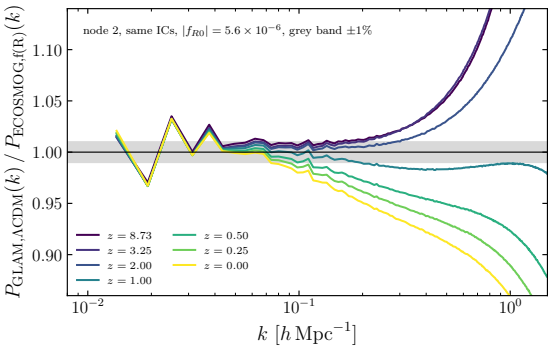
Pilot: node 2 ΛCDM, $N_g = 2048$ (job 12180082)

wall time	2 h 32 min, 64 cores, `cosma8-serial`
CPU efficiency	83 %
peak memory	118 GB (91 GB `PMP2main`)
steps	181, median 0.60 min
outputs	42 $P(k)$, 42 plane sets, 17 catalogues
disk per box	9.1 GB
core-hours	≈ 165 per ΛCDM box

step type	mean wall time
ordinary step	0.60 min
station output: 2048^3 $P(k)$ 25 s, planes 1.8 s, write 12 s	0.96 min
primary output: + BDM halo finder	2.6 min
all 42 outputs	43 min (29 %)
of which lens planes	$< 1\,\%$

- `check_run.py`: schedule, epochs, files, sizes, plane means, catalogues, ic2pm epoch: 0 FAIL.
- BDM dominates the output cost; the new products are cheap. The extras themselves (modes, subsample, f_R) are timed on the f(R) pilot (V9).

Pilot vs the ECOSMOG twin: same ICs, mode-exact



- ECOSMOG has only f(R) spectra, so this is ΛCDM vs the f(R) twin.
- $0.05 < k < 0.3$: a flat $+0.5$ to $+0.7\,\%$ plateau at every epoch, already at $z = 8.7$ where f(R) is negligible.
- Late times: ECOSMOG rises above us by 5 % at $k = 0.1$–0.3 and 10 % at $k = 0.3$–1 by $z = 0$: the f(R) enhancement for $|f_{R0}| = 5.6 \times 10^{-6}$.
- $k > 0.3$ at $z > 3$: +10 % where $P \sim 30\,(h^{-1}\mathrm{Mpc})^3$, the discreteness of the perturbed lattice.

Open: the constant $+0.5\,\%$ at linear scales

Candidates: `ic2pm`'s half-step velocity convention vs ECOSMOG's start, POWMES vs raw-CIC normalisation, a 0.25 % growth difference. The f(R) pilot vs the f(R) spectra decides whether it is a code/ingestion offset; a $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart of 20 steps isolates the half-step term.

f(R) pilot (job 12180110): running since 04:57 BST

- Same node 2 ICs, $|f_{R0}| = 5.6 \times 10^{-6}$, $n = 1$; extras on (δ_m and momentum modes, subsample $M = 100$, f_R at haloes); 400 GB, 16 h limit.
- So far 1.75 min per step, of which the multigrid f(R) solver 1.05 min ($\sim 3\times$ the ΛCDM step); about 7 h in total.
- It will give: V9 (wall-time overhead of the extras, target $< 5\,\%$, and peak memory with the transient $N^3_{\rm dep}$ buffer); the real twin test at linear k (is the $+0.5\,\%$ still there?); subsample and f_R products at 1024^3.
- Caveat: launched with $k_{\rm c,mom} = 1.0$ before the no-interlacing rule, so use its momentum modes below $k = 0.78\,h\,{\rm Mpc}^{-1}$; its binaries predate the review fixes.

Independent review: six reviewers, eight majors, all fixed

review	B	M	m	n
1 Fortran extras module	0	0	9	5
2 scheduler, parameters, build	0	2	6	6
3 particle subsample	0	1	4	6
4 δ_L generator dump	0	0	7	7
5 Python tools and tests	0	2	10	6
6 docs and provenance	0	3	11	8

Self-contained prompts and findings in `docs/reviews/degrace-track-a/`; rerunnable.

Majors and their fixes

- subsample "ID" was the IC file order $\rightarrow$ `ic2pm` scatters by Gadget ID and tags the lattice order
- optional `Init.dat` lines defaulted silently when malformed $\rightarrow$ stop with the line
- f_R gate on `MG_flag` only $\rightarrow$ also `MG_model = 1`
- twin script skipped early epochs (5-decimal header) $\rightarrow$ epochs from the schedule file
- no test sensitive to axis order/sign of the dumps $\rightarrow$ mode-by-mode test
- plan flags table, catalogue velocity epoch, interlacing status $\rightarrow$ documented, rules added

After the fixes: suite 180/180, four scripts green, merged.

Decisions pending and open items

Decisions (yours)

- **BDM catalogue velocities** are converted with $1/a$, the new products with $1/a_v$: a $1.1\,\%$ (first output) to $0.5\,\%$ ($z = 0$) offset. Fix the legacy finder (changes every GLAM catalogue) or keep documenting it (apply a_v/a offline).
- **Momentum deposit** at $N_g = 4096$ f(R): the transient $N^3_{\rm dep}$ buffer (34 GB) does not fit the 985 GB runs; reuse `FI` or restrict extras to $N_g = 2048$.
- **Interlacing**: not implemented; default $k_{c,\rm mom} = 0.75$ stays below $k_{\rm N,dep}/4$ and PMP2init refuses more. Implement if $k > 0.78$ momentum modes are wanted.
- **Disk**: 213 GB regenerable validation IC, 786 GB of February checkpoints in the high-res ΛCDM run, 459 GB in BDM validation work directories, 15 GB of superseded pre-fix fiducial catalogues.

Open minors (grouped in `TRIAGE.md`): parameter-path robustness, f_R product details, provenance JSON, test strength, docs drift. **Questions for Baojiu**: BDM cadence, plane normalisation, CAMB ω_c pitfall, r2g velocity convention, node 900.

Next steps

1. f(R) pilot: `check_run.py`, `compare_ecosmog_pk.py`, V9 overhead and memory; decide on the $+0.5\,\%$ with the $\mathrm{d}a_0$ test if it persists.
2. Settle the three decisions above; rebuild the frozen pilot binaries from `cz`.
3. Production recipe per node: `gen_inputs.py --node N --gravity {lcdm,fr}` $\rightarrow$ `submit.sh` (`ic2pm half` on Baojiu's ICs) $\rightarrow$ `check_run.py`; ≈ 165 core-hours per ΛCDM box, $\approx 3\times$ for f(R); 9 GB + extras (~ 1 GB per output) per box.
4. Readers for the new products in `halocat` (planes, modes, subsample, f_R), following the `.meta` keys.
5. Reply to Baojiu with the cross-check of the 42 epochs and the open questions.

Summary

- The DEGRACE companion outputs (schedule, lens planes, δ_L, modes, subsample, f_R) are implemented, tested, reviewed and merged into `cz`.
- The ΛCDM pilot runs in 2.5 h per box with all 42 outputs; the new products cost about 1 % of the run, the halo finder 23 %.
- Against the ECOSMOG twin the pilot agrees to $+0.5$–$0.7\,\%$ at linear scales at every epoch and shows the expected f(R) signature at late times; the small constant offset is the one physics question still open.
- Three design decisions and the f(R) pilot verification stand between here and production.